

Olist E-Commerce Analytics

Business Problem Statement — Data Analytics Dashboard Project

Project Context

Olist operates a multi-seller e-commerce marketplace across Brazil. Despite steady order volume, leadership lacks a unified view of performance — they see symptoms (revenue swings, late deliveries, uneven seller quality, unpredictable satisfaction scores) but not root causes. This project scopes an end-to-end analytics dashboard so decisions can be made from evidence rather than anecdote.

Each pain point below is documented the way a data analyst would frame it in a project charter: problem statement → business impact → hypothesis / driver → metrics required → decision enabled.

1. Sales Performance

Business Problem

Leadership cannot explain month-to-month revenue swings, has no visibility into which payment channels drive revenue, and cannot identify which regions or periods to prioritize for growth investment.

Business Impact

Without this, marketing spend, inventory buildup, and promotional timing are decided on guesswork — risking wasted budget during low-demand periods and stockouts during peaks.

Working Hypothesis (from initial analysis)

Revenue volatility is driven primarily by order volume, not basket size — AOV stays flat across months, so the November spike and December drop trace back to order counts (likely a Black Friday effect), not pricing or category mix shifts.

Metrics Required

Total revenue, monthly revenue & MoM growth %, AOV trend, revenue by state/city, revenue by payment method.

Decision Enabled

Where to allocate marketing budget, when to plan promotions/inventory buildup, which payment rails to optimize checkout for.

2. Customer Behavior

Business Problem

The business does not know whether growth is coming from new or repeat customers, where its customers are concentrated geographically, or whether satisfaction differs by region.

Business Impact

Blind spot on retention means Olist can't tell if it has a loyalty problem (acquiring but not retaining) versus a pure acquisition-growth story — these require opposite strategic responses.

Metrics Required

Total customers, new vs. repeat customer split, AOV, customer distribution by state/city, average review score, review score by state, purchase trend over time.

Decision Enabled

Whether to invest in retention/loyalty programs vs. acquisition; where to open regional fulfillment or support capacity.

3. Product Performance

Business Problem

Underperforming product categories and high-value categories aren't identified, making inventory planning and category-level investment reactive rather than data-driven.

Business Impact

Overstocking weak categories and understocking strong ones ties up working capital and creates missed-sale risk.

Metrics Required

Best-selling categories, revenue contribution by category, lowest-performing categories, average price by category, order quantity by product, category sales trend.

Decision Enabled

Category-level buy/de-list decisions, pricing adjustments, promotional focus.

4. Seller Performance

Business Problem

Seller quality is inconsistent, some sellers generate a disproportionate share of complaints, and there is no standardized way to rank sellers on revenue, volume, or delivery reliability.

Business Impact

Poor sellers erode marketplace trust and drive down overall satisfaction scores; without visibility, underperformers aren't flagged for review or removal.

Metrics Required

Active seller count, top 10 sellers by revenue/orders, seller distribution by state, average delivery time by seller, seller review ratings, revenue contribution by seller.

Decision Enabled

Seller scorecards, onboarding/offboarding criteria, targeted seller coaching.

5. Order Fulfillment

Business Problem

Order status (completed, cancelled, delivered, pending) isn't systematically monitored, so delay and cancellation patterns go unnoticed until customers complain.

Business Impact

Undetected fulfillment breakdowns compound into customer churn and negative reviews before anyone intervenes.

Metrics Required

Total/completed/cancelled/delivered/pending orders, average delivery days, monthly order trend, order status distribution.

Decision Enabled

Early-warning triggers for operational intervention, capacity planning.

6. Delivery & Logistics

Business Problem

Late deliveries are known to hurt satisfaction, but there's no measurement of where delays originate — by seller, by state, or by product category — so the problem can't be fixed at its source.

Business Impact

Logistics is likely the single biggest lever on customer satisfaction; without diagnosis, fixes are generic rather than targeted.

Metrics Required

Avg shipping days, avg delivery days, late delivery %, delivery performance by seller/state, shipping time by category, shipping trend over time.

Decision Enabled

Carrier/route renegotiation, seller-level SLAs, regional warehousing decisions.

7. Payment Behavior

Business Problem

Finance has no insight into how customers prefer to pay, installment behavior, or how installment plans relate to revenue — limiting checkout optimization and cash flow forecasting.

Business Impact

Cash flow forecasting is harder without understanding installment distribution; checkout friction on preferred payment types can suppress conversion.

Metrics Required

Revenue by payment type, average payment value, installment distribution, average installments, revenue by installment range.

Decision Enabled

Checkout optimization, cash flow planning, fee negotiation with payment processors.

8. Customer Satisfaction

Business Problem

Complaints arrive, but root causes (product category, seller, delivery time) are never connected to review scores, so satisfaction issues get treated symptomatically.

Business Impact

Without linking reviews to their drivers, satisfaction initiatives address the wrong lever (e.g., customer service training when the real issue is late shipping).

Metrics Required

Average review score, score distribution, score by category/seller, delivery days vs. review score correlation, review trend over time.

Decision Enabled

Root-cause-targeted CX improvements rather than blanket fixes.

9. Time Intelligence

Business Problem

Executives ask for trend context (YoY, MoM, running totals) in every review, but there's no standing infrastructure to answer this — it's rebuilt ad hoc each time.

Business Impact

Slower, inconsistent reporting cycles; executives make quarterly decisions without a reliable trend baseline.

Metrics Required

YoY sales, MoM growth, running total, moving average, quarter-wise revenue, weekend vs. weekday orders, monthly customer growth.

Decision Enabled

Standardized executive reporting cadence; faster board/investor updates.

Appendix: Key Figures from Initial Exploratory Analysis

Metric	Value
Total Sales Revenue	R\$16,008,872.12
Total Completed Orders	96,478
Top Revenue State	São Paulo (SP) — R\$2.20M
2nd Revenue State	Rio de Janeiro (RJ) — R\$1.16M
Dominant Payment Method	Credit Card — R\$12.54M (~78%)
2nd Payment Method	Boleto — R\$2.87M (~18%)
Best Performing Month	May — R\$1.75M revenue, highest AOV
Largest MoM Growth	November — +42.36% (Black Friday effect)
Largest MoM Decline	September — -56.83%

Prepared as a data analytics project scoping document. Sections map directly to proposed dashboard pages: Sales, Customers, Products, Sellers, Orders, Logistics, Payments, Satisfaction, and Time Intelligence.